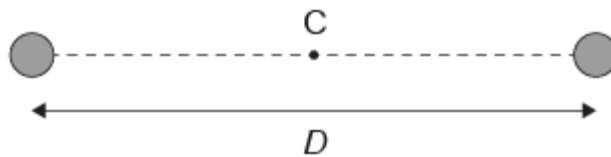


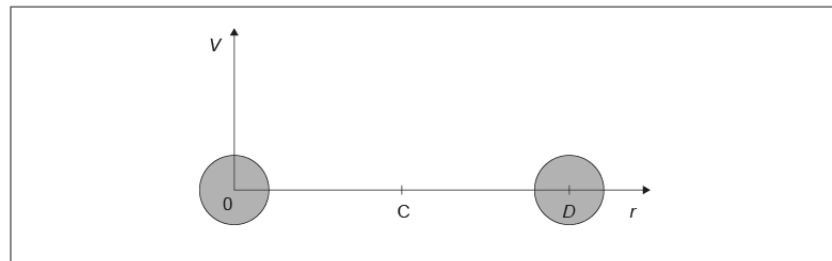
## Topic D HL - Paper 2 - Study Guide [178 marks]

1. [Maximum mark: 11]

- (a) The centres of two identical fixed conducting spheres each of charge  $+Q$  are separated by a distance  $D$ . C is the midpoint of the line joining the centres of the spheres.



- (a.i) Sketch, on the axes, how the electric potential  $V$  due to the two charges varies with the distance  $r$  from the centre of the left charge. No numbers are required. Your graph should extend from  $r = 0$  to  $r = D$ .



[3]

- (a.ii) Calculate the work done to bring a small charge  $q$  from infinity to point C.

Data given:

$$Q = 2.0 \times 10^{-3} \text{ C},$$

$$q = 4.0 \times 10^{-9} \text{ C}$$

$$D = 1.2 \text{ m}$$

[2]

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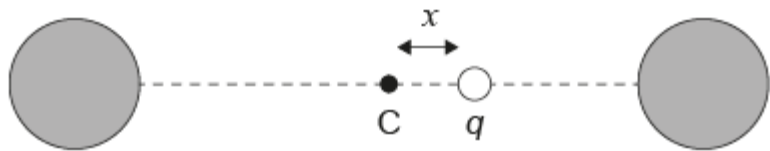
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- (b) The small positive charge  $q$  is placed a distance  $x$  to the right of C. The distance  $x$  is very small compared to  $D$ .



- (b.i) The magnitude of the net force on  $q$  is given by  $\frac{32kQq}{D^3} x$ .  
Explain why the charge  $q$  will execute simple harmonic oscillations about C.

[2]

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- (b.ii) The mass of the charge  $q$  is 0.025 kg.

Calculate the angular frequency of the oscillations using the data in (a)(ii) and the expression in (b)(i).

[2]

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(c) The charges  $Q$  are replaced by neutral masses  $M$  and the charge  $q$  by a neutral mass  $m$ . The mass  $m$  is displaced away from C by a small distance  $x$  and released. Discuss whether the motion of  $m$  will be the same as that of  $q$ .

[2]

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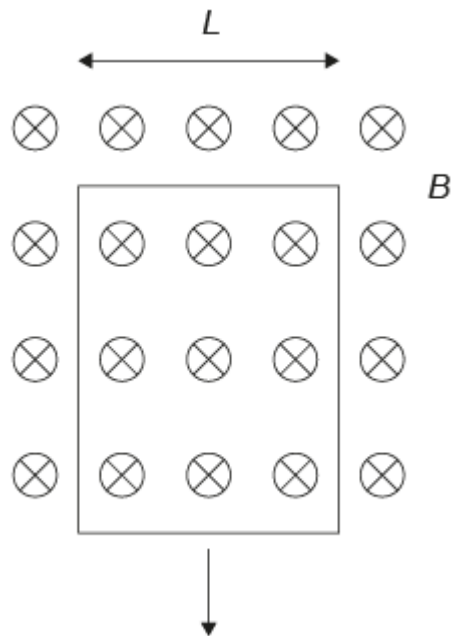
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2. [Maximum mark: 12]

A vertical rectangular loop of conducting wire is dropped in a region of horizontal magnetic field. The diagram shows the loop as it leaves the region of the magnetic field.



- (a) Explain, by reference to Faraday's law of electromagnetic induction, why there is an electromotive force (emf) induced in the loop as it leaves the region of magnetic field.

[2]

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- (b) Just before the loop is about to completely exit the region of magnetic field, the loop moves with constant terminal speed  $v$ .

The following data is available:

|                       |                          |
|-----------------------|--------------------------|
| Mass of loop          | $m = 4.0 \text{ g}$      |
| Resistance of loop    | $R = 25 \text{ m}\Omega$ |
| Width of loop         | $L = 15 \text{ cm}$      |
| Magnetic flux density | $B = 0.80 \text{ T}$     |

Determine, in  $\text{m s}^{-1}$  the terminal speed  $v$ .

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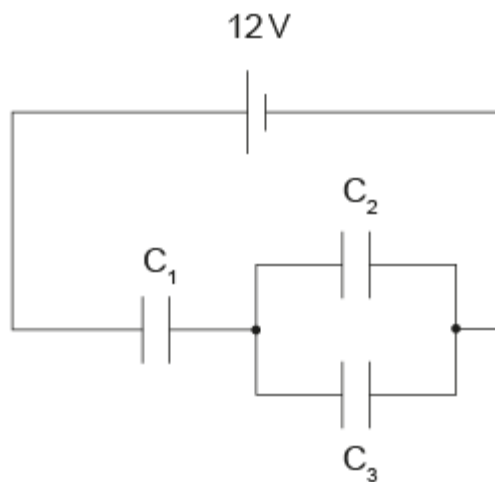
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- (c) Three capacitors  $C_1 = 3.0 \mu\text{F}$ ,  $C_2 = 2.0 \mu\text{F}$  and  $C_3 = 4.0 \mu\text{F}$  are connected to a cell of emf  $12 \text{ V}$  and negligible internal resistance. The capacitors are initially uncharged.



Calculate

- (c.i) the total capacitance of the circuit.

[2]

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(c.ii) the total energy stored in the three capacitors. [1]

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(c.iii) the charge on  $C_2$ . [3]

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**3.** [Maximum mark: 11]

A moon  $M$  orbits a planet  $P$ . The gravitational field strength at the surface of  $P$  due to  $P$  is  $g_P$ .  
 The gravitational field strength at the surface of  $M$  due to  $M$  is  $g_M$ .

For M and P:  $\frac{\text{radius of M}}{\text{radius of P}} = 0.27$  and  $\frac{\text{mass of M}}{\text{mass of P}} = 0.055$

(a) Determine  $\frac{g_M}{g_P}$ .

[2]

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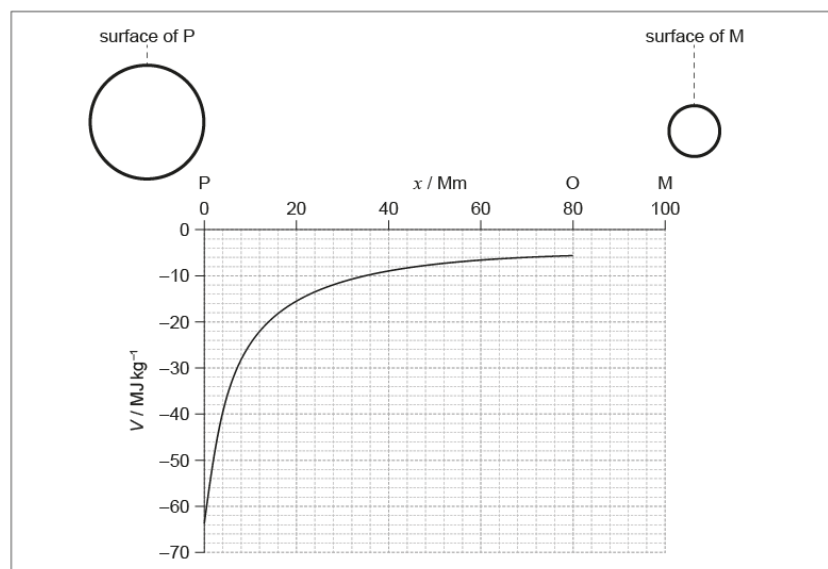
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(b) Point O lies on the line joining the centre of M to the centre of P.



The graph shows the variation of gravitational potential  $V$  with distance  $x$  from the surface of P to O.

The gradient of the graph is zero at point O.

(b.i) State and explain the magnitude of the resultant gravitational field strength at O.

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(b.ii) Outline why the graph between P and O is negative. [2]

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(b.iii) Show that the gravitational potential  $V_P$  at the surface of P due to the mass of P is given by  $V_P = -g_P R_P$  where  $R_P$  is the radius of the planet. [2]

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(b.iv) The gravitational potential due to the mass of M at the surface of P can be assumed to be negligible.

Estimate, using the graph, the gravitational potential at the surface of M due to the mass of M. [2]



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(b.v) Draw on the axes the variation of gravitational potential between O and M. [1]

4. [Maximum mark: 9]  
Two identical positive point charges X and Y are placed 0.30 m apart on a horizontal line. O is the point midway between X and Y. The charge on X and the charge on Y is +4.0  $\mu\text{C}$ .

(a) Calculate the electric potential at O. [3]

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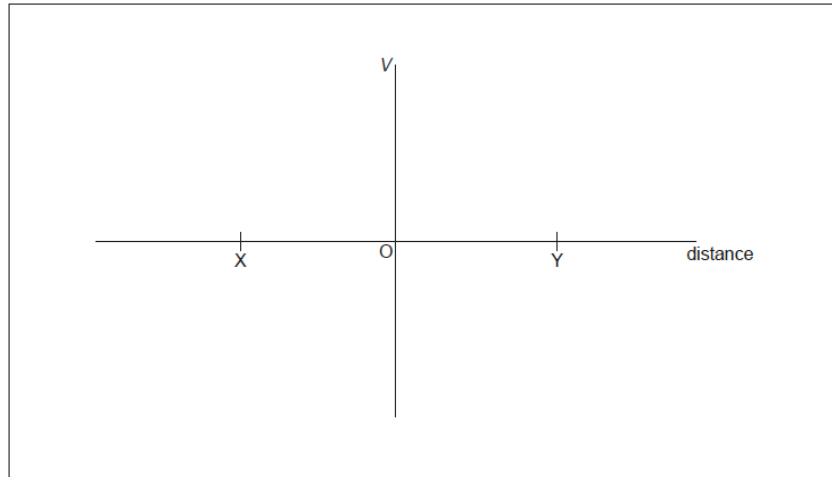
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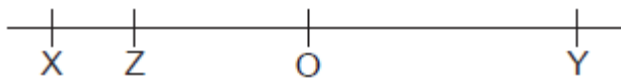
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(b) Sketch, on the axes, the variation of the electric potential  $V$  with distance between X and Y.



[2]

A positive charge Z is released from rest 0.010 m from O on the line between X and Y. Z then begins to oscillate about point O.



(c.i) Identify the direction of the resultant force acting on Z as it oscillates.

[2]

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(c.ii) Deduce whether the motion of Z is simple harmonic.

[2]

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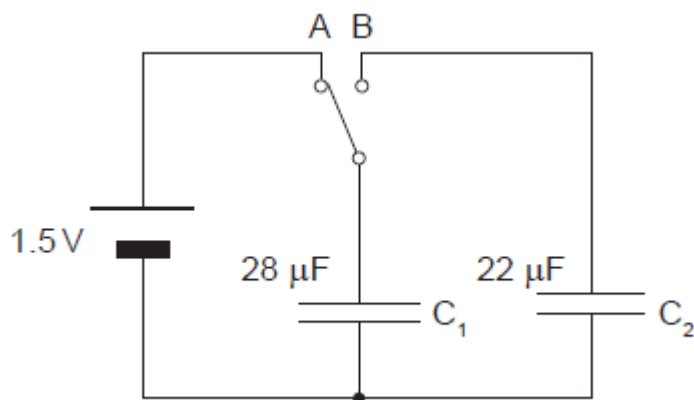
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5. [Maximum mark: 11]

Two capacitors  $C_1$  and  $C_2$  of capacitance  $28\ \mu\text{F}$  and  $22\ \mu\text{F}$  respectively are connected in a circuit with a two-way switch and a cell of emf  $1.5\ \text{V}$  with a negligible internal resistance. The capacitors are initially uncharged. The switch is then connected to position A.



(a) Show that the charge stored on  $C_1$  is about  $0.04\ \text{mC}$ .

[1]

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The switch is moved to position B.

(b.i) Calculate the energy transferred from capacitor  $C_1$ .

[4]

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(b.ii) Explain why the energy gained by capacitor  $C_2$  differs from your answer in (b)(i).

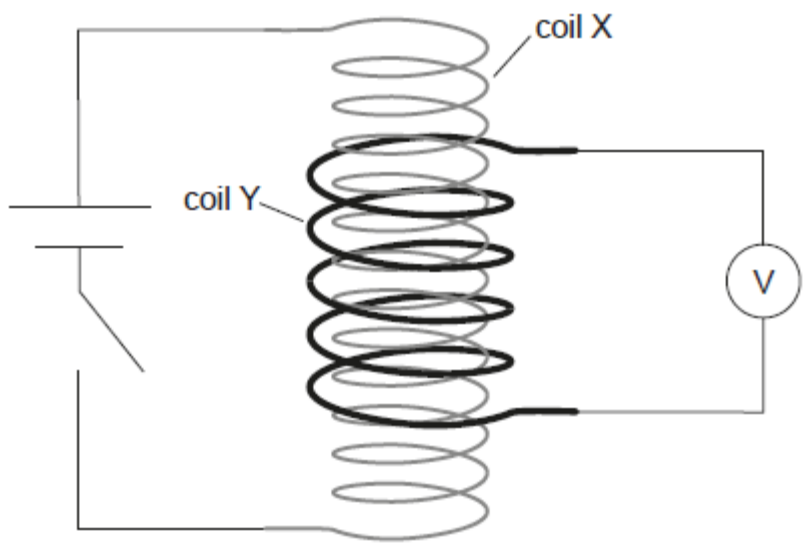
[1]

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A cell is now connected by a switch to a coil X. A second coil Y of cross-sectional area  $6.4 \text{ cm}^2$  with 5 turns is looped around coil X and connected to an ideal voltmeter.



- (c.i) The switch is closed at time  $t = 0$ . Explain how the voltmeter reading varies after the switch is closed.

[2]

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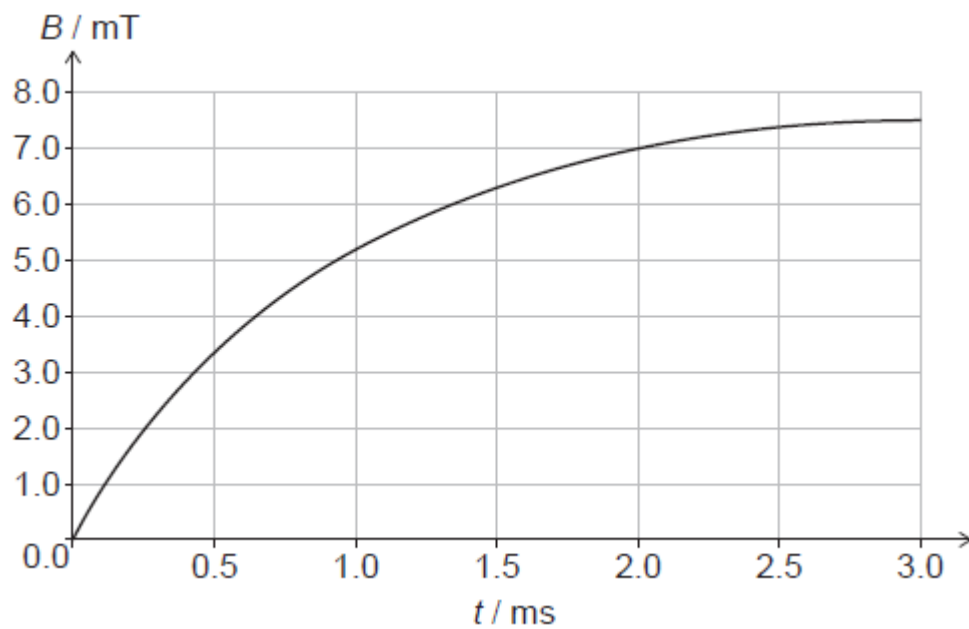
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The graph shows the variation with  $t$  of the magnetic flux density  $B$  in coil Y.



- (c.ii) Determine the average emf induced across coil Y in the first 3.0 ms.

[3]

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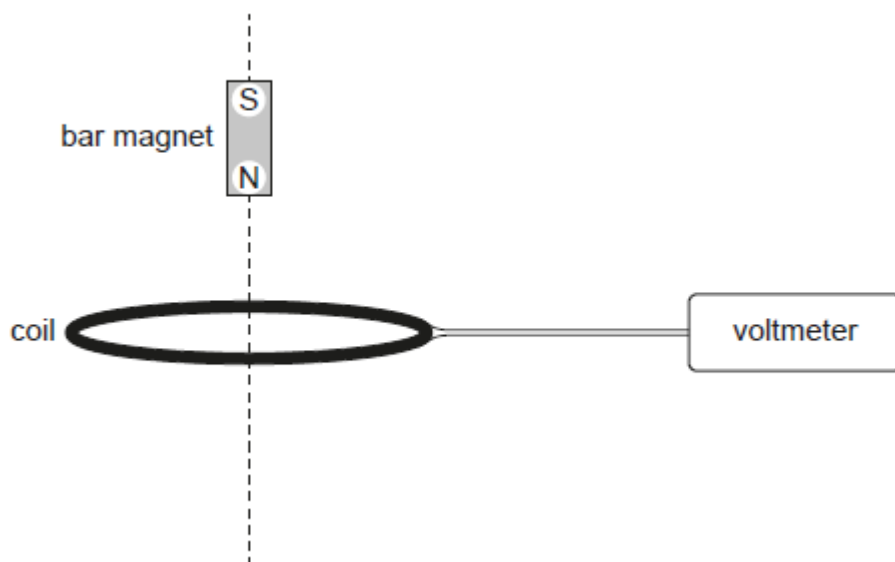
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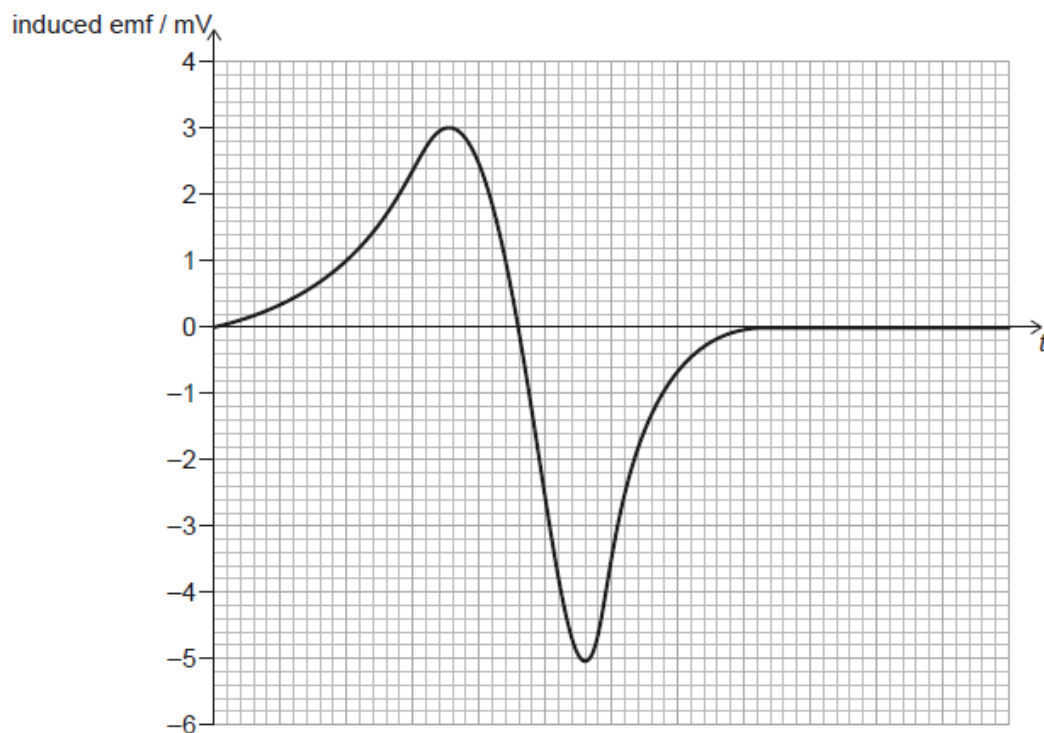
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6. [Maximum mark: 10]

A fixed horizontal coil is connected to an ideal voltmeter. A bar magnet is released from rest so that it falls vertically through the coil along the central axis of the coil.



The variation with time  $t$  of the emf induced in the coil is shown.



- (a.i) Write down the maximum magnitude of the rate of change of flux linked with the coil.

[1]

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- (a.ii) State the fundamental SI unit for your answer to (a)(i).

[1]

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- (b.i) Explain why the graph becomes negative.

[3]

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(b.ii) Part of the graph is above the  $t$ -axis and part is below. Outline why the areas between the  $t$ -axis and the curve for these two parts are likely to be the same.

[2]

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(c) Predict the changes to the graph when the magnet is dropped from a lower height above the coil.

[3]



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7. [Maximum mark: 9]

A metal sphere is charged positively and placed far away from other charged objects. The electric potential at a point on the surface of the sphere is 53.9 kV.

- (a) Outline what is meant by electric potential at a point. [2]

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- (b) The electric potential at a point a distance 2.8 m from the centre of the sphere is 7.71 kV. Determine the radius of the sphere. [2]

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A small positively charged object moves towards the centre of the metal sphere. When the object is 2.8 m from the centre of the sphere, its speed is  $3.1 \text{ m s}^{-1}$ . The mass of the object is 0.14 g and its charge is  $2.4 \times 10^{-8} \text{ C}$ .

(c.i)    Comment on the angle at which the object meets equipotential surfaces around the sphere. [1]

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(c.ii)    Show that the kinetic energy of the object is about 0.7 mJ. [1]

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(c.iii)    Determine whether the object will reach the surface of the sphere. [3]

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8. [Maximum mark: 8]

A planet is in a circular orbit around a star. The speed of the planet is constant.  
 The following data are given:

Mass of planet  $\qquad\qquad\qquad = 8.0 \times 10^{24} \text{ kg}$   
 Mass of star  $\qquad\qquad\qquad = 3.2 \times 10^{30} \text{ kg}$   
 Distance from the star to the planet  $R = 4.4 \times 10^{10} \text{ m}.$

- (a) Explain why a centripetal force is needed for the planet to be in a circular orbit. [2]

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- (b) Calculate the value of the centripetal force. [1]

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A spacecraft is to be launched from the surface of the planet to escape from the star system. The radius of the planet is  $9.1 \times 10^3 \text{ km}$ .

- (c.i) Show that the gravitational potential due to the planet and the star at the surface of the planet is about  $-5 \times 10^9 \text{ J kg}^{-1}$ . [3]

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- (c.ii) Estimate the escape speed of the spacecraft from the planet–star system. [2]

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9. [Maximum mark: 7]

The table gives data for Jupiter and three of its moons, including the radius  $r$  of each object.

| Object   | Mass/kg              | $r / \text{m}$    | Orbital radius<br>around Jupiter/m |
|----------|----------------------|-------------------|------------------------------------|
| Jupiter  | $1.9 \times 10^{27}$ | $7.1 \times 10^7$ |                                    |
| Io       | $8.9 \times 10^{22}$ | $1.8 \times 10^6$ | $4.9 \times 10^8$                  |
| Ganymede | $1.5 \times 10^{23}$ | $2.6 \times 10^6$ | $1.06 \times 10^9$                 |
| Callisto | $1.1 \times 10^{23}$ | $2.4 \times 10^6$ | $1.88 \times 10^9$                 |

- (a) Calculate, for the surface of **Io**, the gravitational field strength  $g_{\text{Io}}$  due to the mass of **Io**. State an appropriate unit for your answer.

[2]

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A spacecraft is to be sent from Io to infinity.

- (b.i) Show that the  $\frac{\text{gravitational potential due to Jupiter at the orbit of Io}}{\text{gravitational potential due to Io at the surface of Io}}$  is about 80.

[2]

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- (b.ii) Outline, using (b)(i), why it is not correct to use the equation

$\sqrt{\frac{2G \times \text{mass of Io}}{\text{radius of Io}}}$  to calculate the speed required for the spacecraft to reach infinity from the surface of Io.

[1]

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- (c) An engineer needs to move a space probe of mass 3600 kg from Ganymede to Callisto. Calculate the energy required to move the probe from the orbital radius of Ganymede to the orbital radius of Callisto. Ignore the mass of the moons in your calculation.

[2]

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**10.** [Maximum mark: 11]

The primary coil of a transformer is connected to a 110 V alternating current (ac) supply. The secondary coil of the transformer is connected to a 15 V garden lighting system that consists of 8 lamps connected in parallel. Each lamp is rated at 35 W when working at its normal brightness. Root mean square (rms) values are used throughout this question.

(a.i) The primary coil has 3300 turns. Calculate the number of turns on the secondary coil.

[1]

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(a.ii) Determine the total resistance of the lamps when they are working normally.

[2]

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(a.iii) Calculate the current in the primary of the transformer assuming that it is ideal.

[2]

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(a.iv) Flux leakage is one reason why a transformer may not be ideal. Explain the effect of flux leakage on the transformer.

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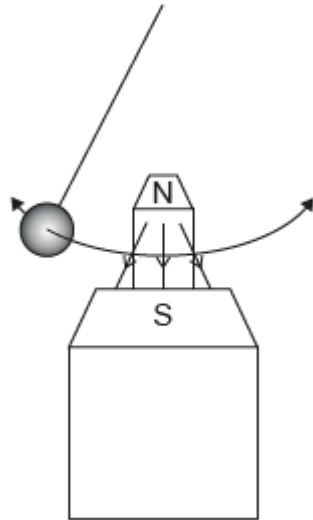
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(b) A pendulum with a metal bob comes to rest after 200 swings. The same pendulum, released from the same position, now swings at  $90^\circ$  to the direction of a strong magnetic field and comes to rest after 20 swings.





[4]

Explain why the pendulum comes to rest after a smaller number of swings.

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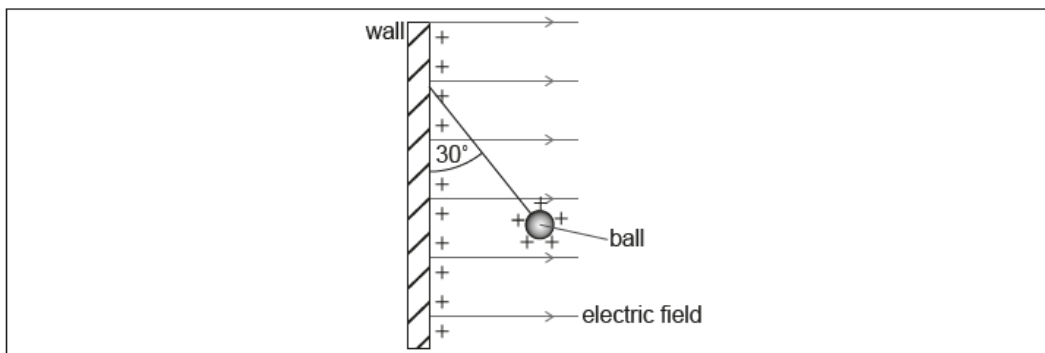
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**11.** [Maximum mark: 15]

A vertical wall carries a uniform positive charge on its surface. This produces a uniform horizontal electric field perpendicular to the wall. A small, positively-charged ball is suspended in equilibrium from the vertical wall by a thread of negligible mass.



- (a) The charge per unit area on the surface of the wall is  $\sigma$ . It can be shown that the electric field strength  $E$  due to the charge on the wall is given by the equation

$$E = \frac{\sigma}{2\epsilon_0}.$$

Demonstrate that the units of the quantities in this equation are consistent.

[2]

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- (b.i) The thread makes an angle of 30° with the vertical wall. The ball has a mass of 0.025 kg.

Determine the horizontal force that acts on the ball.

[3]

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(b.ii) The charge on the ball is  $1.2 \times 10^{-6} \text{ C}$ . Determine  $\sigma$ . [2]

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(c) The thread breaks. Explain the initial subsequent motion of the ball. [3]

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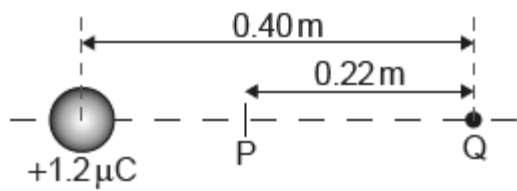
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The centre of the ball, still carrying a charge of  $1.2 \times 10^{-6} \text{ C}$ , is now placed 0.40 m from a point charge Q. The charge on the ball acts as a point charge at the centre of the ball.

P is the point on the line joining the charges where the electric field strength is zero. The distance PQ is 0.22 m.



(d.i) Calculate the charge on Q. State your answer to an appropriate number of significant figures.

[3]

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- (d.ii) Outline, without calculation, whether or not the electric potential at P is zero.

[2]

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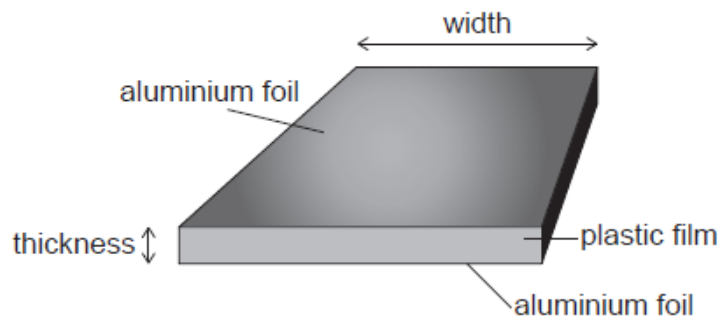
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**12.** [Maximum mark: 12]

A student makes a parallel-plate capacitor of capacitance 68 nF from aluminium foil and plastic film by inserting one sheet of plastic film between two sheets of aluminium foil.

**diagram not to scale**



The aluminium foil and the plastic film are 450 mm wide.

The plastic film has a thickness of 55  $\mu\text{m}$  and a permittivity of  $2.5 \times 10^{-11} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$ .

- (a.i) Calculate the total length of aluminium foil that the student will require.

[3]

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(a.ii) The plastic film begins to conduct when the electric field strength in it exceeds  $1.5 \text{ MN C}^{-1}$ . Calculate the maximum charge that can be stored on the capacitor.

[2]

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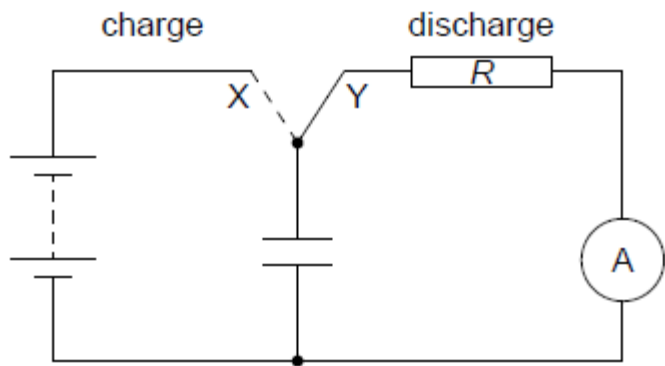
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The student uses a switch to charge and discharge the capacitor using the circuit shown. The ammeter is ideal.



The emf of the battery is 12 V.

- (b.i) The resistor  $R$  in the circuit has a resistance of  $1.2\text{ k}\Omega$ . Calculate the time taken for the charge on the capacitor to fall to 50 % of its fully charged value.

[3]

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- (b.ii) The ammeter is replaced by a coil. Explain why there will be an induced emf in the coil while the capacitor is discharging.

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- (b.iii) Suggest **one** change to the discharge circuit, apart from changes to the coil, that will increase the maximum induced emf in the coil.

[2]

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**13.** [Maximum mark: 10]

The moon Phobos moves around the planet Mars in a circular orbit.

(a.i) Outline the origin of the force that acts on Phobos. [1]

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(a.ii) Outline why this force does no work on Phobos. [1]

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(b.i) The orbital period  $T$  of a moon orbiting a planet of mass  $M$  is given by

$$\frac{R^3}{T^2} = kM$$

where  $R$  is the average distance between the centre of the planet and the centre of the moon.

Show that  $k = \frac{G}{4\pi^2}$  [3]



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(b.ii) The following data for the Mars–Phobos system and the Earth–Moon system are available:

Mass of Earth =  $5.97 \times 10^{24}$  kg

The Earth–Moon distance is 41 times the Mars–Phobos distance.

The orbital period of the Moon is 86 times the orbital period of Phobos.

Calculate, in kg, the mass of Mars.

[2]

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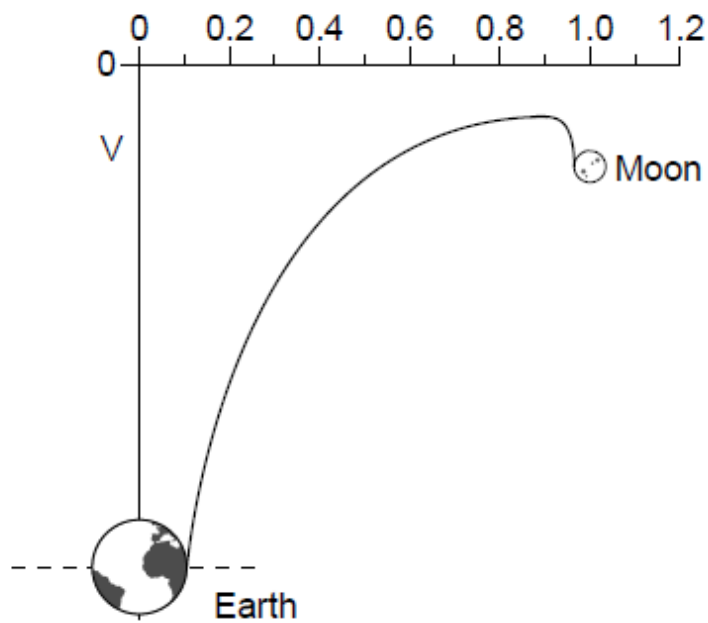
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(c) The graph shows the variation of the gravitational potential between the Earth and Moon with distance from the centre of the Earth. The distance from the Earth is expressed as a fraction

of the total distance between the centre of the Earth and the centre of the Moon.



[3]

Determine, using the graph, the mass of the Moon.

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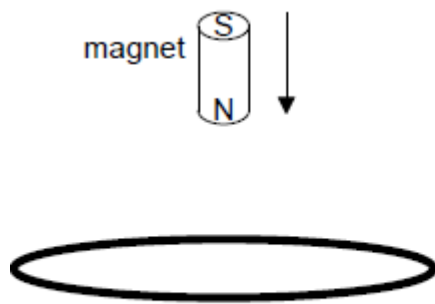
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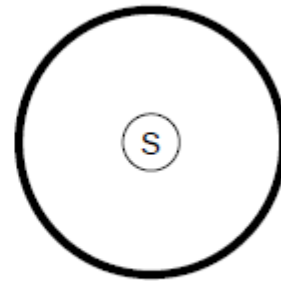
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14. [Maximum mark: 4]

A small magnet is dropped from rest above a stationary horizontal conducting ring. The south (S) pole of the magnet is upwards.



**Diagram 1: side view**



**Diagram 2: view from above**

- (a) While the magnet is moving towards the ring, state why the magnetic flux in the ring is increasing.

[1]



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- (b) While the magnet is moving towards the ring, sketch, using an arrow on **Diagram 2**, the direction of the induced current in the ring.

[1]



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- (c) While the magnet is moving towards the ring, deduce the direction of the magnetic force on the magnet.

[2]



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15. [Maximum mark: 6]

A planet of mass  $m$  is in a circular orbit around a star. The gravitational potential due to the star at the position of the planet is  $V$ .

- (ai) Show that the total energy of the planet is given by the equation shown.

$$E = \frac{1}{2}mV$$

[2] 

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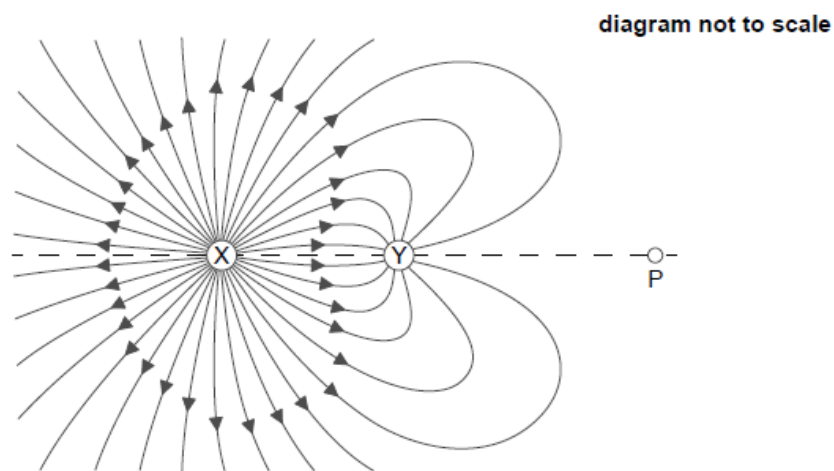
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- (aii) Suppose the star could contract to half its original radius without any loss of mass. Discuss the effect, if any, this has on the total energy of the planet.

[2] 

- (b) The diagram shows some of the electric field lines for two fixed, charged particles X and Y.



The magnitude of the charge on X is  $Q$  and that on Y is  $q$ . The distance between X and Y is 0.600 m. The distance between P and Y is 0.820 m.

At P the electric field is zero. Determine, to **one** significant figure, the ratio  $\frac{Q}{q}$ .

[2]



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16. [Maximum mark: 4]  
 A small magnet is dropped from rest above a stationary horizontal conducting ring. The south (S) pole of the magnet is upwards.

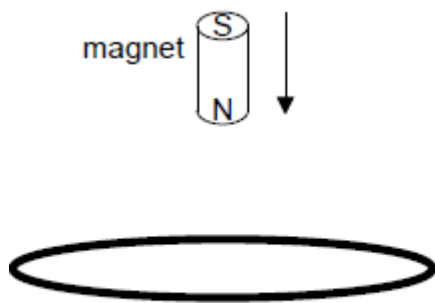


Diagram 1: side view

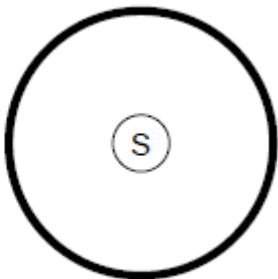


Diagram 2: view from above

- (a) While the magnet is moving towards the ring, state why the magnetic flux in the ring is increasing.

[1] □

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- (b) While the magnet is moving towards the ring, sketch, using an arrow on **Diagram 2**, the direction of the induced current in the ring.

[1] □

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(c) While the magnet is moving towards the ring, deduce the direction of the magnetic force on the magnet. [2] 

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**17.** [Maximum mark: 6]  
A planet of mass  $m$  is in a circular orbit around a star. The gravitational potential due to the star at the position of the planet is  $V$ .

(ai) Show that the total energy of the planet is given by the equation shown.

$$E = \frac{1}{2}mV$$

[2] 

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(aii) Suppose the star could contract to half its original radius without any loss of mass. Discuss the effect, if any, this has on the total energy of the planet.

[2] 

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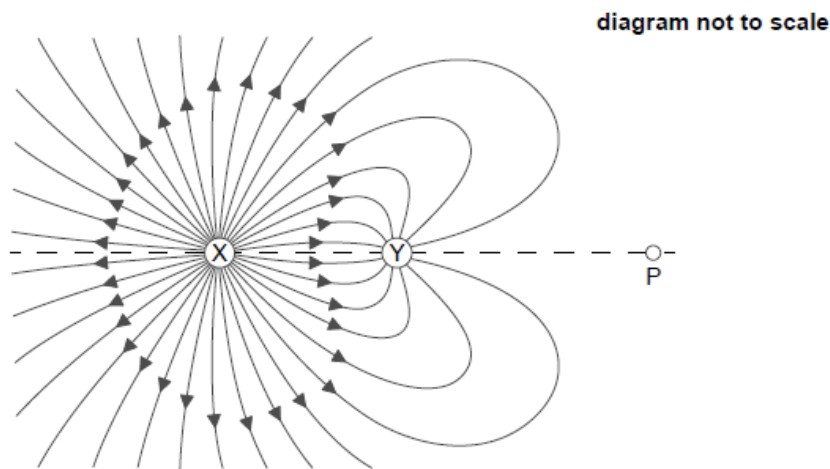
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(b) The diagram shows some of the electric field lines for two fixed, charged particles X and Y.



The magnitude of the charge on X is  $Q$  and that on Y is  $q$ . The distance between X and Y is 0.600 m. The distance between P



and Y is 0.820 m.

At P the electric field is zero. Determine, to **one** significant figure, the ratio  $\frac{Q}{q}$ .

[2] 

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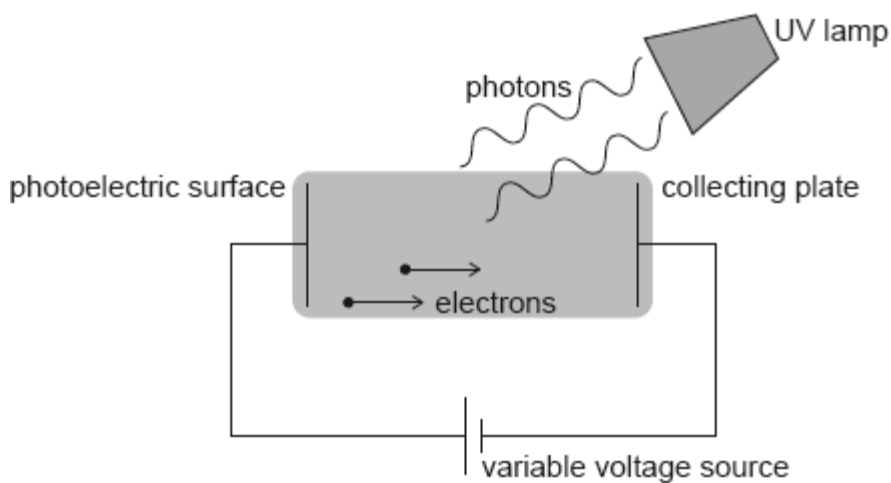
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**18.** [Maximum mark: 11]

Hydrogen atoms in an ultraviolet (UV) lamp make transitions from the first excited state to the ground state. Photons are emitted and are incident on a photoelectric surface as shown.



(a) Show that the energy of photons from the UV lamp is about 10 eV.

[2]

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The photons cause the emission of electrons from the photoelectric surface. The work function of the photoelectric surface is 5.1 eV.

(b.i) Calculate, in J, the maximum kinetic energy of the emitted electrons. [2]

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(b.ii) Suggest, with reference to conservation of energy, how the variable voltage source can be used to stop all emitted electrons from reaching the collecting plate. [2]

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- (b.iii) The variable voltage can be adjusted so that no electrons reach the collecting plate. Write down the minimum value of the voltage for which no electrons reach the collecting plate.

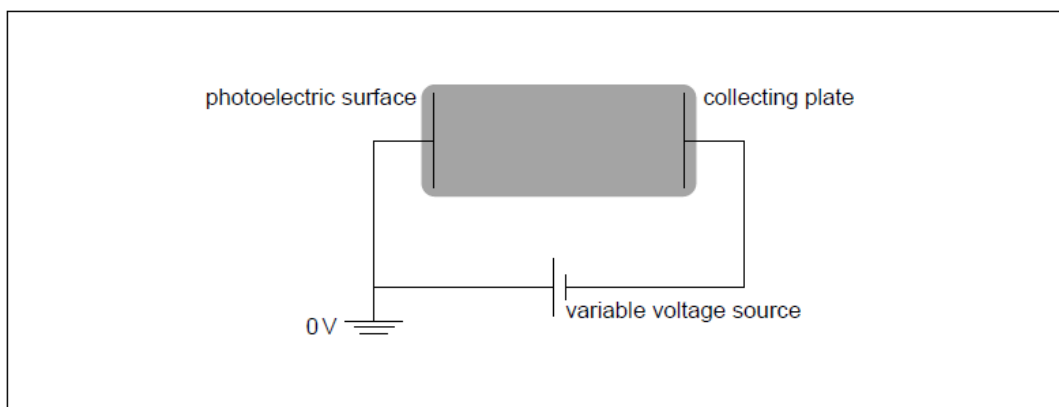
[1]

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The electric potential of the photoelectric surface is 0V. The variable voltage is adjusted so that the collecting plate is at  $-1.2\text{V}$ .



- (c.i) On the diagram, draw and label the equipotential lines at  $-0.4\text{V}$  and  $-0.8\text{V}$ .

[2]

- (c.ii) An electron is emitted from the photoelectric surface with kinetic energy  $2.1\text{ eV}$ . Calculate the speed of the electron at the collecting plate.

[2]

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19. [Maximum mark: 11]

A planet has radius  $R$ . At a distance  $h$  above the surface of the planet the gravitational field strength is  $g$  and the gravitational potential is  $V$ .

(a.i) State what is meant by gravitational field strength.

[1]

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(a.ii) Show that  $V = -g(R + h)$ .

[2]

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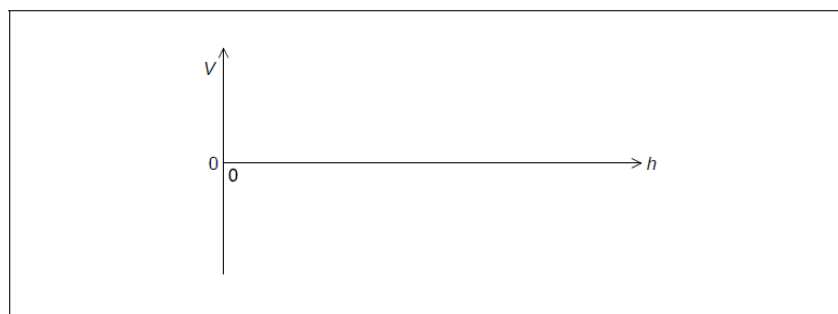
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(a.iii) Draw a graph, on the axes, to show the variation of the gravitational potential  $V$  of the planet with height  $h$  above the surface of the planet.



[2]

(b) A planet has a radius of  $3.1 \times 10^6$  m. At a point P a distance  $2.4 \times 10^7$  m above the surface of the planet the gravitational field

strength is  $2.2 \text{ N kg}^{-1}$ . Calculate the gravitational potential at point P, include an appropriate unit for your answer.

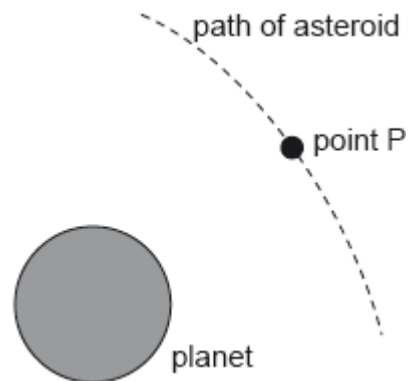
[1]

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- (c) The diagram shows the path of an asteroid as it moves past the planet.



When the asteroid was far away from the planet it had negligible speed. Estimate the speed of the asteroid at point P as defined in (b).

[3]

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- (d) The mass of the asteroid is  $6.2 \times 10^{12} \text{ kg}$ . Calculate the gravitational force experienced by the **planet** when the asteroid is at point P.

[2]

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